

Module -4

Applications of Laplace Transform for Circuits without Initial conditions

Dr. Sathya P
Associate Professor
SENSE

Circuit Element Models

- **Steps in applying the Laplace Transform:**
- Transform the circuit from time domain to the s-domain
- Solve the circuit using nodal analysis, mesh analysis, source transformation, superposition, or any other circuit analysis technique with which we are familiar.
- Take the inverse transform of the solution and thus obtain the solution in the time domain.

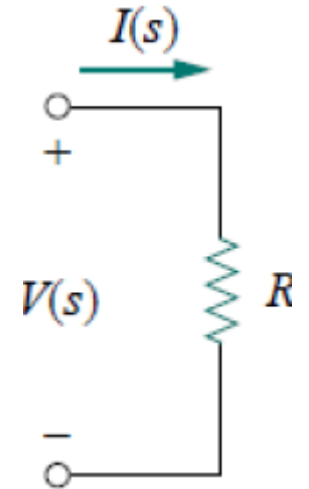
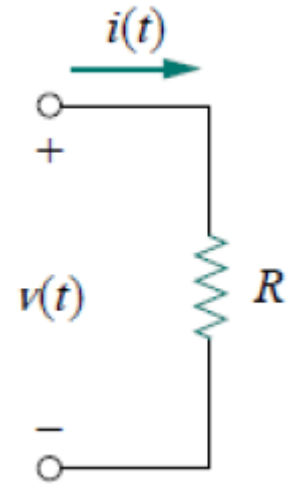
Resistor

For a resistor, the voltage-current relationship in the time domain is

$$v(t) = Ri(t)$$

Taking the Laplace transform, we get

$$V(s) = RI(s)$$



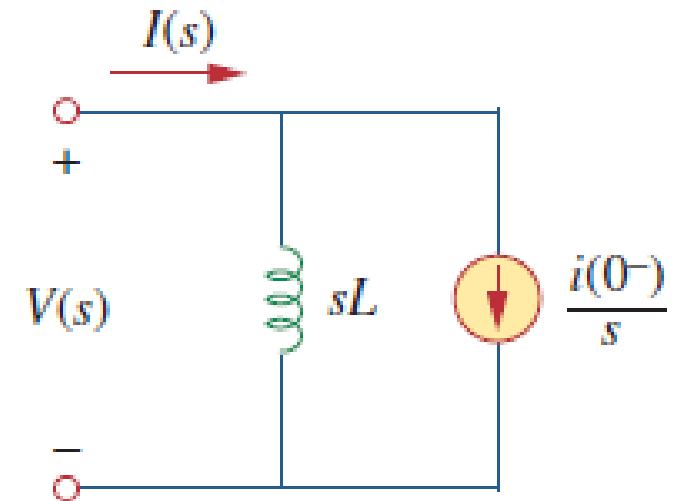
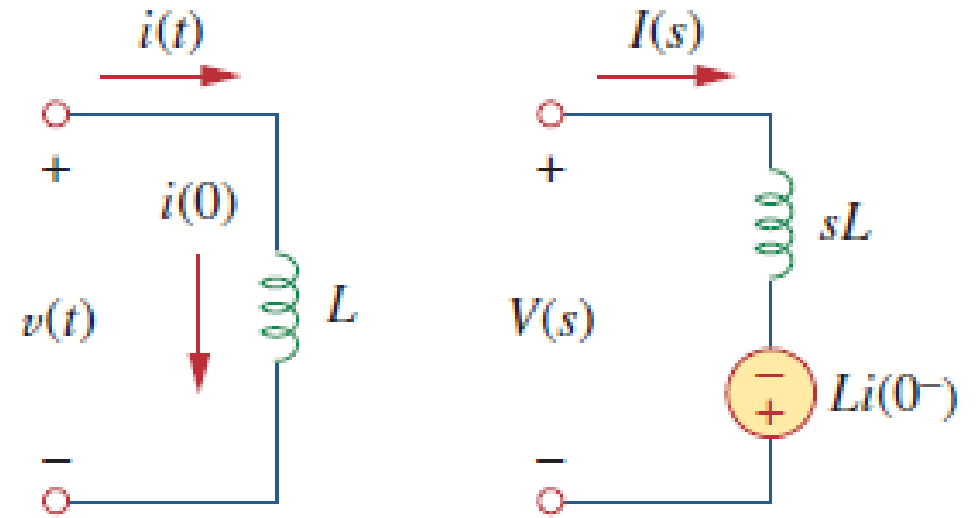
Inductor

$$v(t) = L \frac{di(t)}{dt}$$

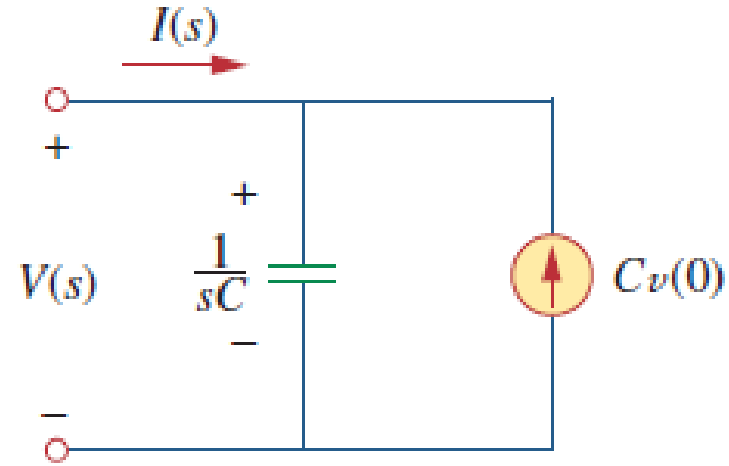
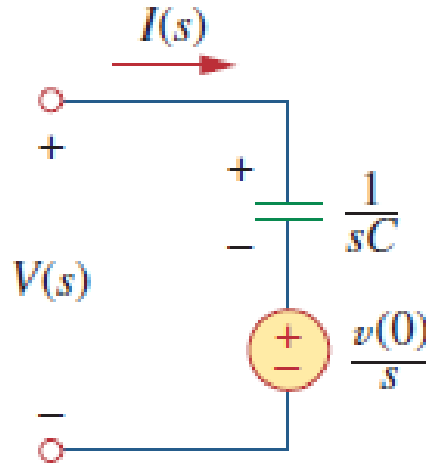
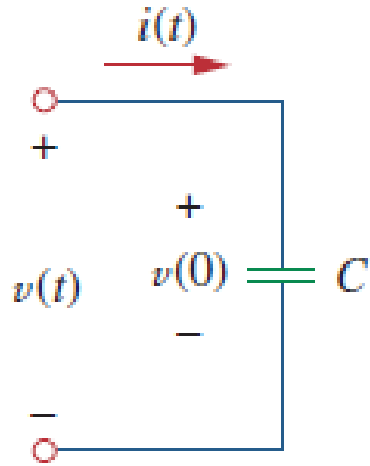
Taking the Laplace transform of both sides

$$V(s) = L[sI(s) - i(0^-)] = sLI(s) - Li(0^-)$$

$$I(s) = \frac{1}{sL} V(s) + \frac{i(0^-)}{s}$$



Capacitor

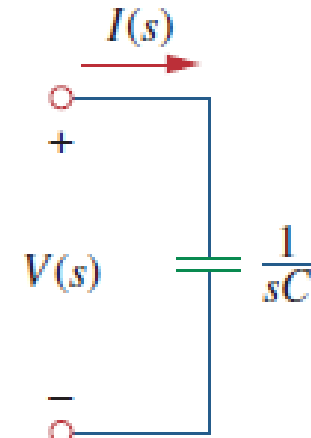
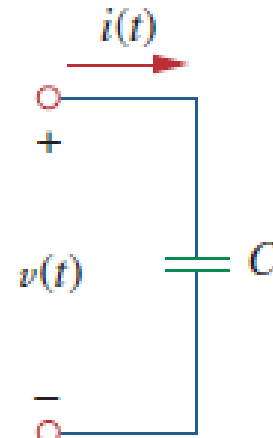
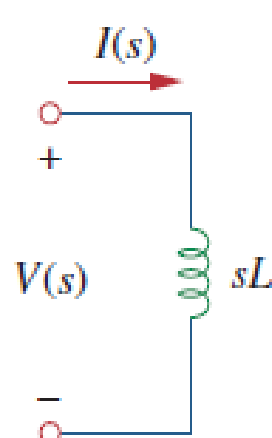
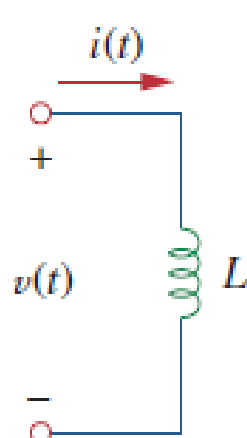
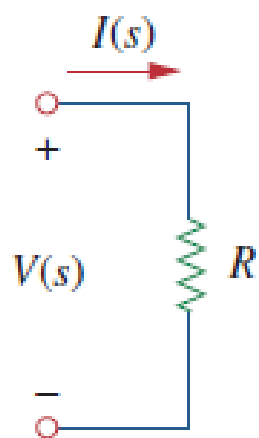
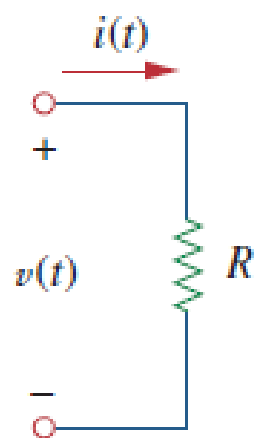


$$i(t) = C \frac{dv(t)}{dt}$$

$$I(s) = C[sV(s) - v(0^-)] = sCV(s) - Cv(0^-)$$

$$V(s) = \frac{1}{sC}I(s) + \frac{v(0^-)}{s}$$

Circuit Elements with Zero initial conditions



Resistor: $V(s) = RI(s)$

Inductor: $V(s) = sLI(s)$

Capacitor: $V(s) = \frac{1}{sC}I(s)$

Element

$Z(s) = V(s)/I(s)$

Resistor

R

Inductor

sL

Capacitor

$1/sC$

Numerical - 1

Find $v_o(t)$ in the circuit, assuming zero initial conditions.

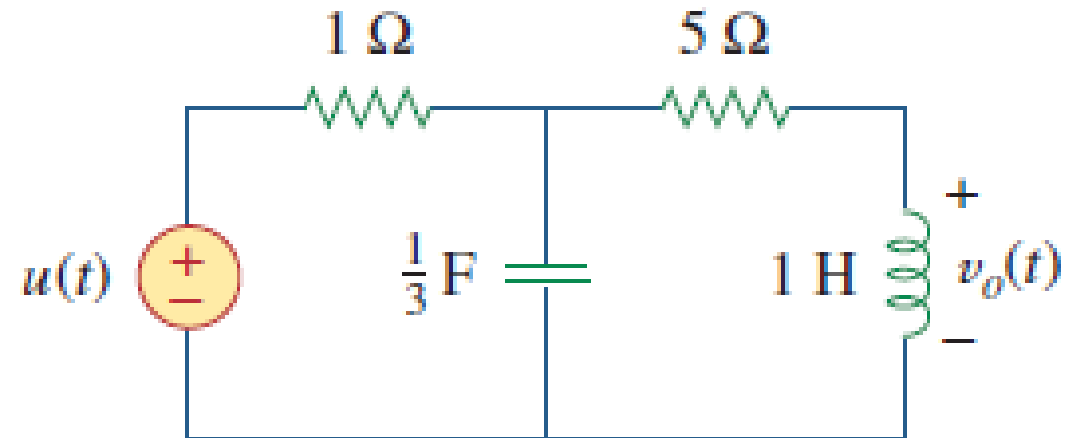
Solution:

time domain to the s -domain.

$$u(t) \Rightarrow \frac{1}{s}$$

$$1 \text{ H} \Rightarrow sL = s$$

$$\frac{1}{3} \text{ F} \Rightarrow \frac{1}{sC} = \frac{3}{s}$$



For mesh 1,

$$\frac{1}{s} = \left(1 + \frac{3}{s}\right)I_1 - \frac{3}{s}I_2 \longrightarrow (1)$$

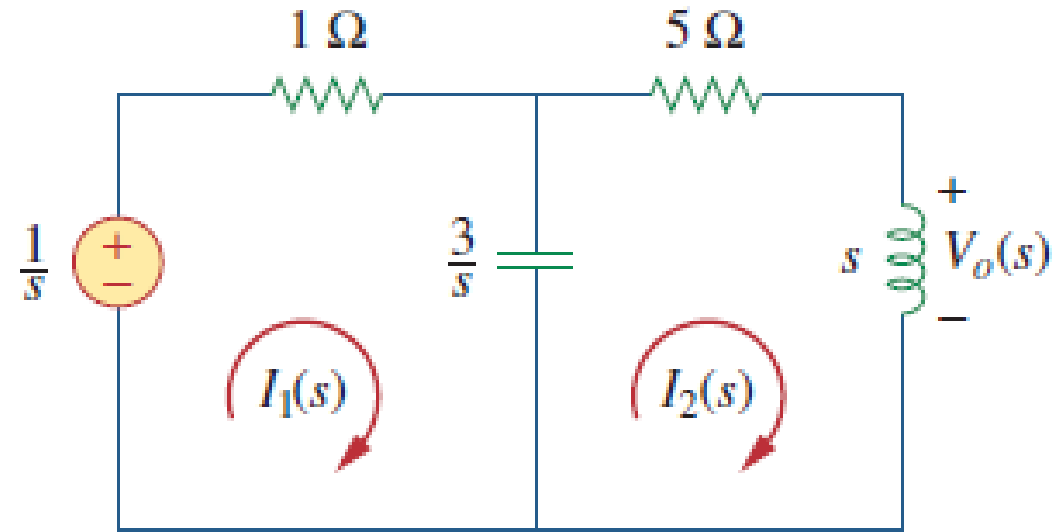
For mesh 2,

$$0 = -\frac{3}{s}I_1 + \left(s + 5 + \frac{3}{s}\right)I_2$$

$$I_1 = \frac{1}{3}(s^2 + 5s + 3)I_2 \longrightarrow (2)$$

Sub, eqn (2) in (1)

$$\frac{1}{s} = \left(1 + \frac{3}{s}\right)\frac{1}{3}(s^2 + 5s + 3)I_2 - \frac{3}{s}I_2$$



Multiplying through by $3s$ gives

$$3 = (s^3 + 8s^2 + 18s)I_2 \quad \Rightarrow \quad I_2 = \frac{3}{s^3 + 8s^2 + 18s}$$

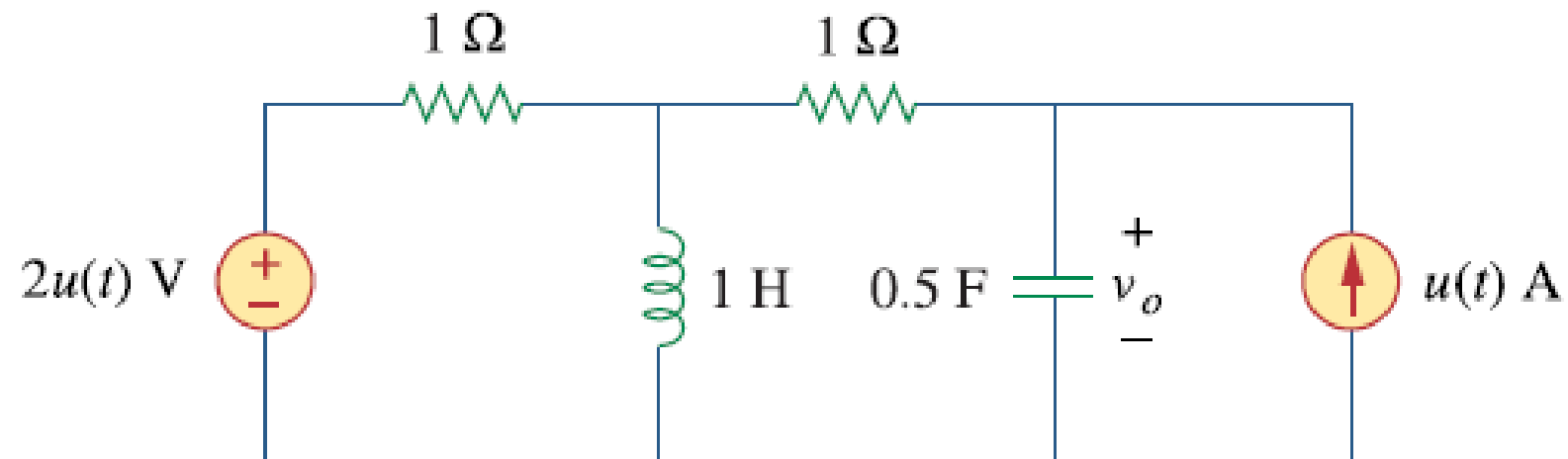
$$V_o(s) = sI_2 = \frac{3}{s^2 + 8s + 18} = \frac{3}{\sqrt{2}} \frac{\sqrt{2}}{(s + 4)^2 + (\sqrt{2})^2}$$

Taking the inverse transform yields

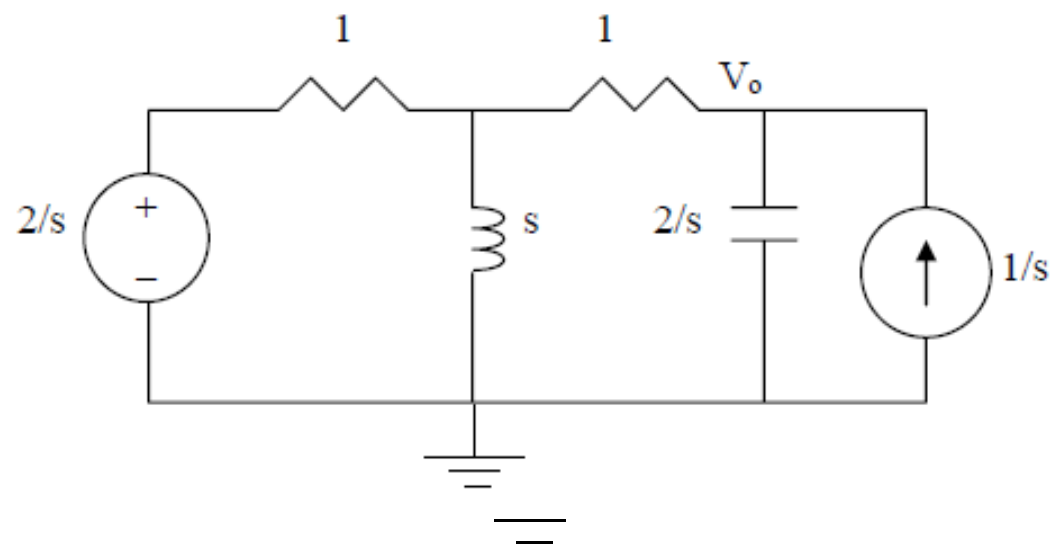
$$v_o(t) = \frac{3}{\sqrt{2}} e^{-4t} \sin \sqrt{2}t \text{ V}, \quad t \geq 0$$

Numerical - 2

Find $v_o(t)$, for all $t > 0$, in the circuit



Solution



At node 1

$$\frac{2/s - V_1}{1} = \frac{V_1}{s} + \frac{V_1 - V_o}{1} \longrightarrow \frac{2}{s} = V_1(2 + 1/s) - V_o \quad (1)$$

At node O,

$$\frac{V_1 - V_o}{1} + \frac{1}{s} = \frac{V_o}{2/s} = \frac{s}{2}V_o \longrightarrow V_1 = (1 + s/2)V_o - 1/s \quad (2)$$

Substituting (2) into (1) gives

$$2/s = (2 + 1/s)(1 + s/2)V_o - \frac{1}{s}(2 + \frac{1}{s}) - V_o$$

Substituting (2) into (1) gives

$$2/s = (2 + 1/s)(1 + s/2)V_o - \frac{1}{s}(2 + \frac{1}{s}) - V_o$$

$$V_o = \frac{(4s+1)}{s(s^2+1.5s+1)} = \frac{A}{s} + \frac{Bs+C}{s^2+1.5s+1}$$

$$4s+1 = A(s^2+1.5s+1) + Bs^2 + Cs$$

We equate coefficients.

$$s^2 : \quad 0 = A + B \quad \text{or} \quad B = -A$$

$$s : \quad 4 = 1.5A + C$$

$$\text{constant:} \quad 1 = A, \quad B = -1, \quad C = 4 - 1.5A = 2.5$$

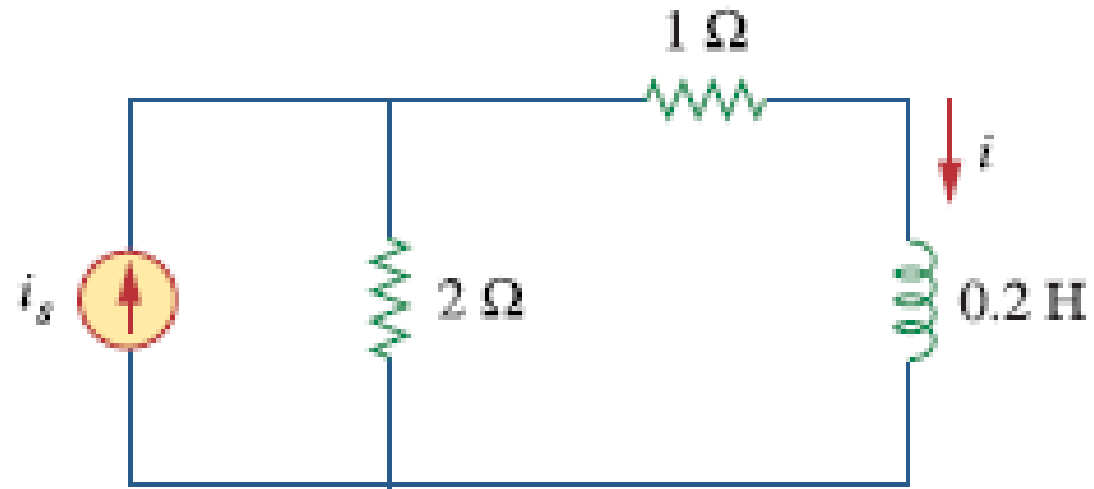
$$V_o = \frac{1}{s} + \frac{-s+2.5}{s^2+1.5s+1} = \frac{1}{s} - \frac{s+3/4}{(s+3/4)^2 + \left(\frac{\sqrt{7}}{4}\right)^2} + \frac{\frac{3.25}{\sqrt{7}} \times \frac{\sqrt{7}}{4}}{(s+3/4)^2 + \left(\frac{\sqrt{7}}{4}\right)^2}$$

$$\underline{v(t) = u(t) - e^{-3t/4} \cos \frac{\sqrt{7}}{4} t + 4.9135 e^{-3t/4} \sin \frac{\sqrt{7}}{4} t}$$

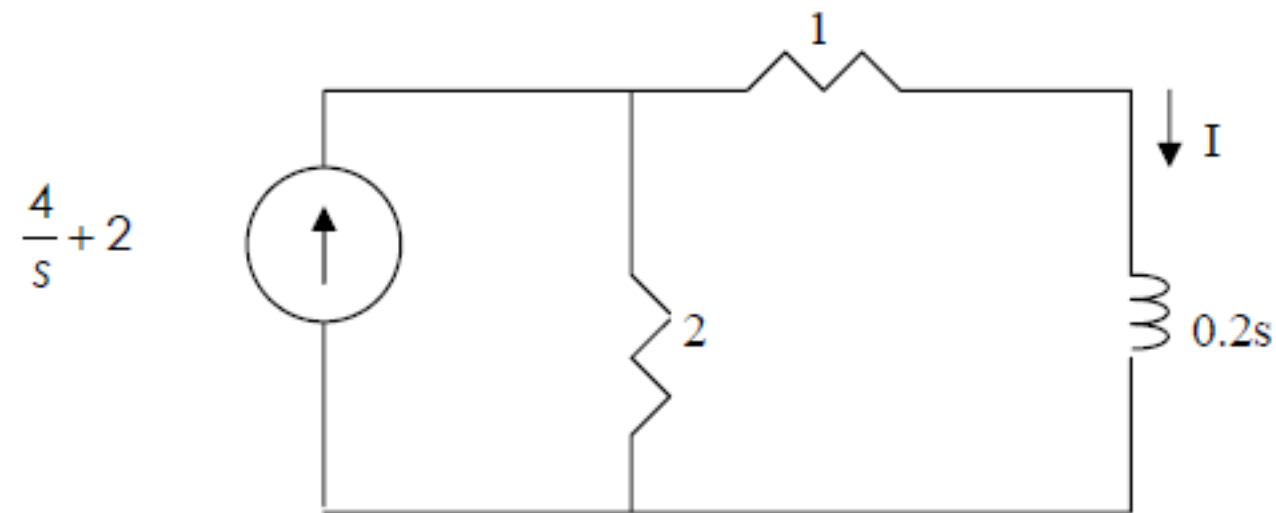
Numerical - 3

Find $i(t)$ for $t > 0$ for the circuit

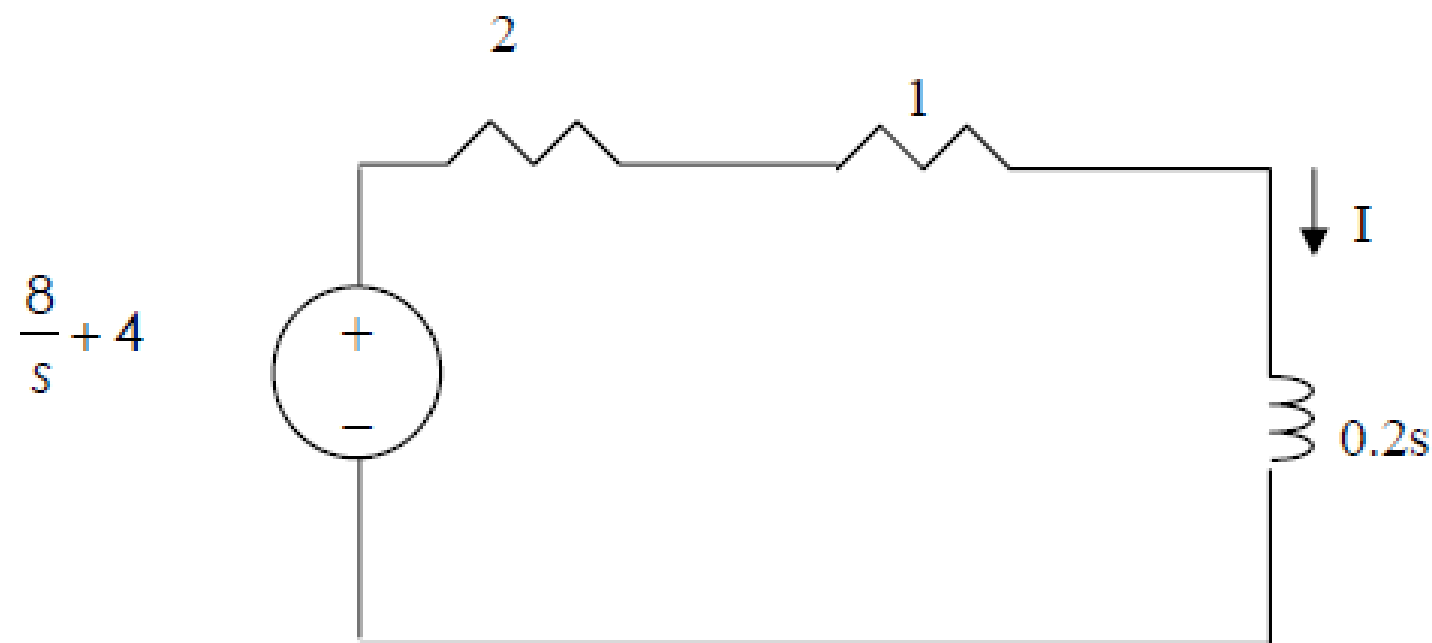
Assume $i_s = 4u(t) + 2\delta(t)$ mA.



Solution



We transform the current source to a voltage source and obtain the circuit shown below.



$$I = \frac{\frac{8}{s} + 4}{3 + 0.2s} = \frac{20s + 40}{s(s + 15)} = \frac{A}{s} + \frac{B}{s + 15}$$

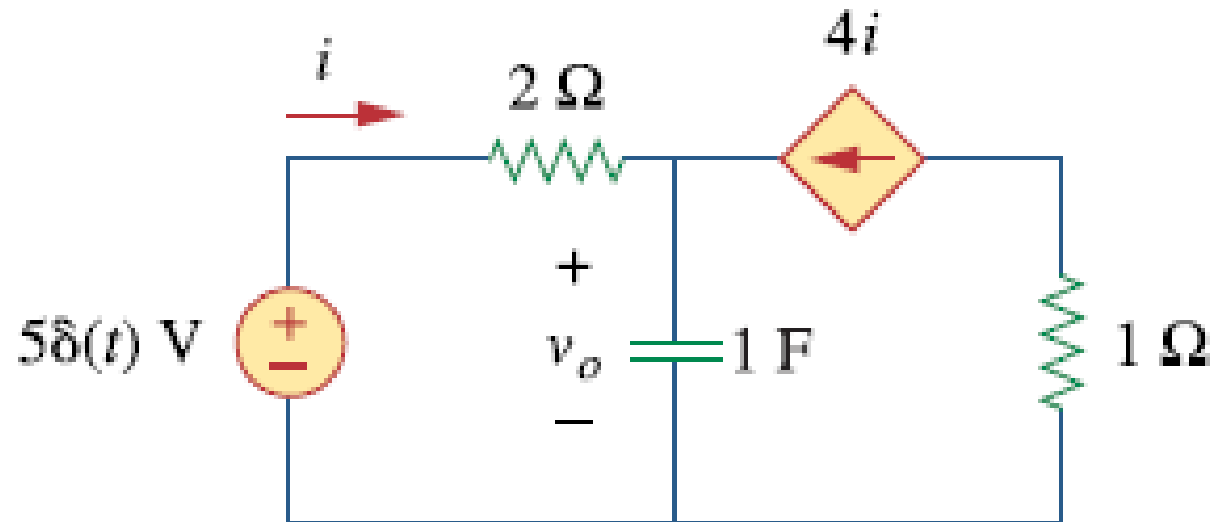
$$A = \frac{40}{15} = \frac{8}{3}, \quad B = \frac{-15 \times 20 + 40}{-15} = \frac{52}{3}$$

$$I = \frac{8/3}{s} + \frac{52/3}{s + 15}$$

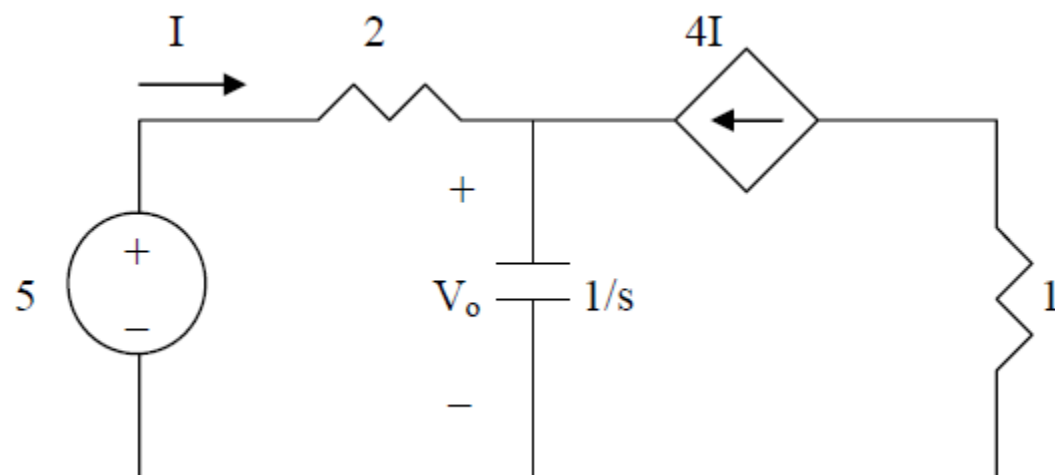
$$\underline{i(t) = \left[\frac{8}{3} + \frac{52}{3} e^{-15t} \right] u(t)}$$

Numerical - 4

The capacitor in the circuit of Fig. is initially uncharged. Find $v_o(t)$ for $t > 0$.



Solution



$$I + 4I = \frac{V_o}{1/s} \longrightarrow 5I = sV_o$$

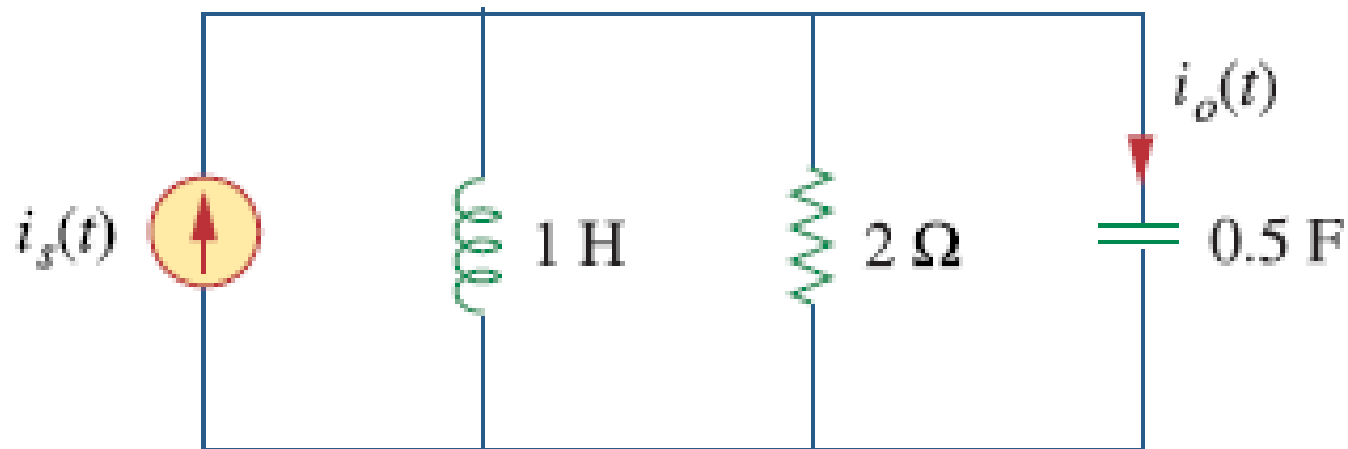
$$\text{But } I = \frac{5 - V_o}{2}$$

$$5\left(\frac{5 - V_o}{2}\right) = sV_o \longrightarrow V_o = \frac{12.5}{s + 5/2}$$

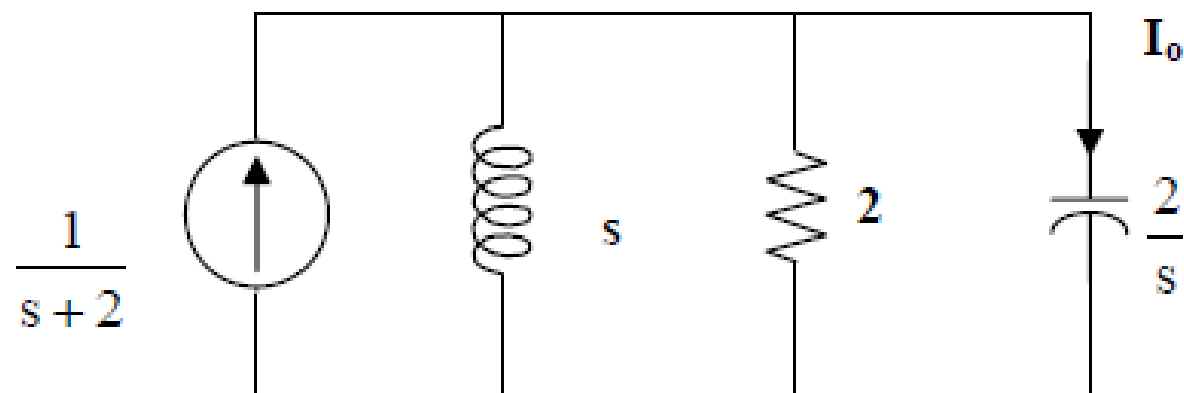
$$v_o(t) = \underline{12.5e^{-2.5t} \text{ V}}$$

Numerical - 5

If $i_s(t) = e^{-2t} u(t)$ A in the circuit
find the value of $i_o(t)$.



Solution



$$V = \frac{1}{s+2} \left(\frac{1}{\frac{1}{s} + \frac{1}{2} + \frac{s}{2}} \right) = \frac{1}{s+2} \left(\frac{2s}{s^2 + s + 2} \right) = \frac{2s}{(s+2)(s+0.5+j1.3229)(s+0.5-j1.3229)}$$

$$I_o = \frac{Vs}{2} = \frac{s^2}{(s+2)(s+0.5+j1.3229)(s+0.5-j1.3229)}$$

$$= \frac{1}{s+2} + \frac{\frac{(-0.5-j1.3229)^2}{(1.5-j1.3229)(-j2.646)}}{s+0.5+j1.3229} + \frac{\frac{(-0.5+j1.3229)^2}{(1.5+j1.3229)(+j2.646)}}{s+0.5-j1.3229}$$

$$i_o(t) = \left(e^{-2t} + 0.3779e^{-90^\circ} e^{-t/2} e^{-j1.3229t} + 0.3779e^{90^\circ} e^{-t/2} e^{j1.3229t} \right) u(t) \text{ A}$$

$$= \left(e^{-2t} - 0.7559 \sin 1.3229t \right) u(t) \text{ A}$$